

# Ibrahim Khebour

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## Education

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| <b>Colorado State University</b><br><i>PhD in Computer Science</i>                                              | Aug 2022 – Present<br>Fort Collins, CO |
| <b>Tunisia Polytechnic School</b><br><i>Multidisciplinary Engineering Degree</i>                                | Sep 2019 – Jun 2022<br>Tunis, Tunisia  |
| <b>Preparatory Institute for Engineering Studies of El Manar</b><br><i>Ranked 36<sup>th</sup>/2000 (Top 2%)</i> | Sep 2017 – Jun 2019<br>Tunis, Tunisia  |

## Research Experience

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| <b>Graduate Research Assistant</b><br><i>Colorado State University, SIGNAL Lab</i>                                                                                                                                                                                                                                               | Aug 2022 – Present<br>Fort Collins, CO  |
| <ul style="list-style-type: none"><li>– Architecting multimodal ML systems combining speech, vision, and behavioral signals for common ground tracking.</li><li>– Designing real-time inference pipelines for low-latency deployment in live environments.</li><li>– Contributing to NSF iSAT and DARPA FACT projects.</li></ul> |                                         |
| <b>Graduation Research Intern</b><br><i>Colorado State University</i>                                                                                                                                                                                                                                                            | Feb 2022 – Jun 2022<br>Fort Collins, CO |
| <ul style="list-style-type: none"><li>– Developed multilingual loanword detection models across 10+ languages.</li><li>– Applied data augmentation for low-resource NLP tasks.</li></ul>                                                                                                                                         |                                         |
| <b>Data Science Intern</b><br><i>Datagram</i>                                                                                                                                                                                                                                                                                    | Jun 2021 – Aug 2021<br>Paris, France    |
| <ul style="list-style-type: none"><li>– Built semantic similarity systems using Transformer models (BERT, RoBERTa, USE).</li></ul>                                                                                                                                                                                               |                                         |

## Publications

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- **Khebour, Ibrahim**, Jung, Changsoo, Fitzgerald, Jack, Jamil, Huma, Krishnaswamy, Nikhil. *Feature Contributions to Multimodal Interpretation of Meaning*. Proceedings of the International Conference on Human-Computer Interaction (HCII), 2025.
- Jamil, Huma, **Khebour, Ibrahim**, Lai, Kenneth, Pustejovsky, James, Krishnaswamy, Nikhil. *A Graph Autoencoder Approach for Gesture Classification with Gesture AMR*. Proceedings of the International Conference on Computational Semantics (IWCS), 2025.
- Zhu, Yifan, Lai, Kenneth, **Khebour, Ibrahim**, Verhagen, Marc, Krishnaswamy, Nikhil, Pustejovsky, James. *Multimodal Situational Awareness: Neuro-Symbolic AI for Real-Time HCI in the Classroom*. Proceedings of the International Conference on Human-Computer Interaction (HCII), 2025.
- Bradford, Mariah, **Khebour, Ibrahim**, VanderHoeven, Hannah, Venkatesha, Videep, Blanchard, Nathaniel, Krishnaswamy, Nikhil. *Tracking Individual Beliefs in Co-situated Groups Using Multimodal Input*. Proceedings of the International Conference on Human-Computer Interaction (HCII), 2025.

- Venkatesha, Videep, Nath, Abhijnan, **Khebour, Ibrahim**, Chelle, Avyakta, Bradford, Mariah, Tu, Jingxuan, VanderHoeven, Hannah, Bhalla, Brady, Youngren, Austin, Fitzgerald, Jack, et al. *Propositional Extraction from Collaborative Naturalistic Dialogues*. Journal of Educational Data Mining, 2025.
- VanderHoeven, Hannah, Bhalla, Brady, **Khebour, Ibrahim**, Youngren, Austin, Venkatesha, Videep, Bradford, Mariah, Fitzgerald, Jack, Mabrey, Carlos, Tu, Jingxuan, Zhu, Yifan, et al. *TRACE: Real-Time Multimodal Common Ground Tracking in Situated Collaborative Dialogues*. Proceedings of NAACL-HLT (System Demonstrations), 2025.
- Palmer, Derek, Zhu, Yifan, Lai, Kenneth, VanderHoeven, Hannah, Bradford, Mariah, **Khebour, Ibrahim**, Mabrey, Carlos, Fitzgerald, Jack, Krishnaswamy, Nikhil, Palmer, Martha, et al. *Speech Is Not Enough: Interpreting Nonverbal Indicators of Common Knowledge and Engagement*. Proceedings of the AAAI Conference on Artificial Intelligence, 2025.
- VanderHoeven, Hannah, Bradford, Mariah, Jung, Changsoo, **Khebour, Ibrahim**, Lai, Kenneth, Pustejovsky, James, Krishnaswamy, Nikhil, Blanchard, Nathaniel. *Multimodal Design for Interactive Collaborative Problem-Solving Support*. Proceedings of the International Conference on Human-Computer Interaction (HCII), 2024.
- Zhu, Yifan, VanderHoeven, Hannah, Lai, Kenneth, Bradford, Mariah, Tam, Christopher, **Khebour, Ibrahim**, Brutti, Richard, Krishnaswamy, Nikhil, Pustejovsky, James. *Modeling Theory of Mind in Multimodal HCI*. Proceedings of the International Conference on Human-Computer Interaction (HCII), 2024.
- Venkatesha, Videep, Nath, Abhijnan, **Khebour, Ibrahim**, Chelle, Avyakta, Bradford, Mariah, Tu, Jingxuan, Pustejovsky, James, Blanchard, Nathaniel, Krishnaswamy, Nikhil. *Propositional Extraction from Natural Speech in Small Group Collaborative Tasks*. International Educational Data Mining Society, 2024.
- **Khebour, Ibrahim**, Lai, Kenneth, Bradford, Mariah, Zhu, Yifan, Brutti, Richard, Tam, Christopher, Tu, Jingxuan, Ibarra, Benjamin, Blanchard, Nathaniel, Krishnaswamy, Nikhil, et al. *Common Ground Tracking in Multimodal Dialogue*. Proceedings of LREC-COLING, 2024.
- **Khebour, Ibrahim**, Brutti, Richard, Dey, Indrani, Dickler, Rachel, Sikes, Kelsey, Lai, Kenneth, Bradford, Mariah, Cates, Brittany, Hansen, Paige, Jung, Changsoo, et al. *When Text and Speech Are Not Enough: A Multimodal Dataset of Collaboration in a Situated Task*. Journal of Open Humanities Data, 2024.
- Terpstra, Corbyn, **Khebour, Ibrahim**, Bradford, Mariah, Wisniewski, Brett, Krishnaswamy, Nikhil, Blanchard, Nathaniel. *How Good is Automatic Segmentation as a Multimodal Discourse Annotation Aid?* Proceedings of the Joint ACL-ISO Workshop on Interoperable Semantics (ISA), 2023.
- Bradford, Mariah, **Khebour, Ibrahim**, Blanchard, Nathaniel, Krishnaswamy, Nikhil. *Automatic Detection of Collaborative States in Small Groups Using Multimodal Features*. Proceedings of the International Conference on Artificial Intelligence in Education (AIED), 2023.
- Nath, Abhijnan, Saravani, Sina Mahdipour, **Khebour, Ibrahim**, Mannan, Sheikh, Li, Zihui, Krishnaswamy, Nikhil. *A Generalized Method for Automated Multilingual Loanword Detection*. Proceedings of the International Conference on Computational Linguistics (COLING), 2022.

### *Specialized Skills*

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- **Programming:** Python (PyTorch, HuggingFace), C/C++, R, SQL, LaTeX
- **Core Areas:** Multimodal ML, NLP, Real-Time Systems
- **Mathematics:** Linear Algebra, Probability, Optimization
- **Languages:** English, French, Arabic